Local–Global Spectral Feature-Aware Learning for Hyperspectral Imagery Classification

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***Abstract*—Effective modeling of the relationship between local spectral details and global contextual information remains a core challenge in hyperspectral image (HSI) classification. In this article, a local–global spectral feature-aware network (LGSFA-Net) is proposed, which achieves local and global spectral feature learning through synergistic integration of local convolutional inductive biases and global state-space models (SSMs). The architecture of LGSFA-Net comprises three sequential components, including an embedding stage using convolutions for fundamental feature extraction, an encoding stage that cascades standard Mamba blocks with specialized interactive Mamba (IMamba) blocks, and an enhanced spatial–spectral feature fusion (ESSFF) module. The proposed IMamba blocks employ separable convolutions and feature interaction learning for explicitly modeling the cross-channel spectral correlations learning, which can be effective in awareness of the spectral feature. And then, the ESSFF module utilizes self-attention mechanisms to dynamically balance local and global spatial–spectral feature contributions. The final prediction stage incorporates a lightweight classification head for efficient inference. Experimental results validate the effectiveness of the proposed methods for HSI classification on four benchmark datasets, including the Pavia University (PaviaU), Houston, Honghu, and Hanchuan datasets. The proposed LGSFA-Net achieves approximately 1.48%–2.51% increased overall accuracy (OA), 1.34%–2.06% increased average accuracy (AA), and 1.37%–3.75% increased Kappa on the aforementioned four datasets, respectively, outperforming the contrasting methods. The code implementation will be available at: https://github.com/yutinyang/LGSFA-Net.**

***Index Terms*—Channel feature, feature fusion, hyperspectral image (HSI) classification, Mamba, spectral feature learning.**

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I. Introduction

REMOTE sensing (RS) images parsing holds significant research significance and practical value in fields such as environmental monitoring, agriculture, and mineral exploration [1], [2]. Hyperspectral images (HSIs) in RS often provide rich spectral information, aiding in identifying subtle differences in land cover features. In HSI interpretation, deep learning is crucial because it can automatically extract complex features, enhancing the accuracy and efficiency of image interpretation [3].

In recent years, convolutional neural networks (CNNs) and Transformer frameworks in deep learning have taken the dominant position in RS image interpretation [4], [5]. In 2023, Mamba [6] emerged and garnered widespread attention from researchers due to its outstanding global modeling capabilities and linear computational complexity superior to Transformer [7]. Initially proposed for natural language processing tasks, it was later extended to visual representation learning tasks and widely applied in various fields, including medicine, RS, and others [8].

With the development of RS image interpretation methods, various approaches have emerged, such as RSMamba [9], HyperMamba [10], IGroupSS-Mamba [11], and MLMamba [12]. According to the framework type, these Mamba-based methods can be classified into pure Mamba-based RS image interpretation models and hybrid Mamba-based models that combine CNNs, Transformers, and Mamba for RS image interpretation. For instance, models like RSDehamba [13] for haze removal in RS satellite images, RSM [14] for dense prediction tasks in very high-resolution (VHR) RS images, and RSMamba [9] for RS image classification are pure Mamba-based frameworks. On the other hand, methods like DualMamba [15], RS3Mamba [16], MamTrans [17], and RTMamba [18] belong to the hybrid Mamba framework for RS image interpretation. In detail, DualMamba [15] combines CNN modules with the Mamba framework for HSI classification, effectively extracting global and local spectral spatial features. RS3Mamba [16] constructs a RS semantic segmentation network by combining visual state-space (VSS) blocks and CNN's residual blocks. MamTrans [17] combines the Transformer and Mamba frameworks, effectively utilizing the Transformer's ability to capture spatial token relationships and Mamba's ability to extract implicit features along tokens. RTMamba [18] integrates convolutional blocks, Mamba's VSS

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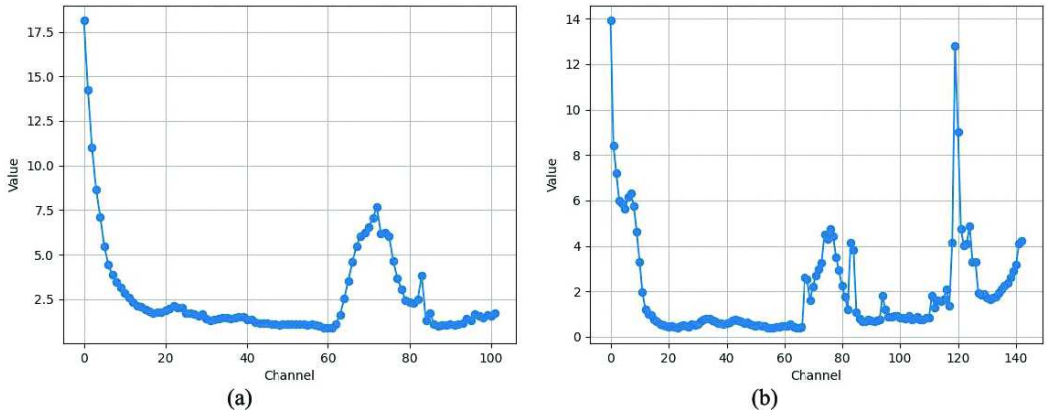


Fig. 1. Examples of average channel difference on PaviaU and Houston datasets, including channel average difference curve on (a) PaviaU dataset and (b) Houston dataset.

blocks, and a semantic Transformer to build a real-time RS semantic segmentation model. In addition, all the hybrid framework models mentioned above can leverage the strengths of CNNs, Transformer, and Mamba to achieve more robust feature representation.

In addition, spatial–spectral feature learning still plays an important role in HSI classification [19]. Most of the Mamba-based work focuses on spatial–spectral feature learning to enhance Mamba's spatial–spectral feature learning capabilities, such as SS-Mamba [20], 3DSS-Mamba [21], SSUMamba [22], S^2Mamba [23], and MambaHSI [24]. However, apart from spectral features, HSI images are often multichannel. The image information contained in different channels varies. As shown in Fig. 1, the channel average values are curved about the Pavia University (PaviaU) and Houston datasets. The curves demonstrate that different channels have different average values, indicating varying feature information across channels that could impact the final recognition results differently. This highlights the importance of channelwise feature learning in enhancing the performance of the Mamba model in HSI classification tasks. Furthermore, the local correlation between pixels often plays a crucial role in HSI recognition and classification. Studies including SLCGC [25] and AHSGC [26] adopt GCN as their backbone or feature extractor to perform global relational modeling via graph structures. In a word, effectively modeling local feature correlation will significantly improve the performance of pixel classification in HSI.

Based on the aforementioned foundations, a hybrid Mamba framework is proposed for HSI classification. While Mamba primarily focuses on modeling remote global dependencies, local and channel features are crucial for HSI's pixel-level classification. To enhance HSI's local feature and channelwise spectral feature learning, the interactive Mamba (IMamba) block is introduced, which combines a local encoder and a channel encoder to model local and channel features. It emphasizes the relationships between pixel-level features within an image, considering channel differences.

Unlike existing hybrid Mamba frameworks, the proposed local–global spectral feature-aware network (LGSFA-Net) presents a local–global spectral feature-aware learning for HSI classification, rather than a simplistic combination of convolution and Mamba framework. In detail, the introduction of an IMamba block that explicitly learns cross-channel spectral correlations by modeling pixelwise interactions; the synergistic coupling of local convolutional feature extraction with global state-space modeling, enabling unified local–global spectral feature learning; and the use of an adaptive fusion module to dynamically optimize feature contributions. It adaptively merges Mamba and IMamba features while addressing the varying impacts of these features on classification performance. Experiments conducted on four benchmark HSI classification datasets demonstrate outstanding accuracy, yielding impressive performance metrics such as overall accuracy (OA), average accuracy (AA), and Kappa (k). In summary, the main contributions of this article are shown as follows.

1) The hybrid Mamba-based LGSFA-Net network is proposed, which introduces local convolution and the global Mamba block to achieve good spectral feature representation learning for better HSI classification.
2) The IMamba block is presented, which incorporates spectral feature correlation learning (FCL) based on Mamba, including local and channel spectral FCL.
3) An enhanced spatial–spectral feature fusion (ESSFF) module is proposed, which utilizes a self-attention mechanism to dynamically balance and adaptively integrate the local and global spatial–spectral features extracted by the standard Mamba block and the IMamba block, thereby effectively leveraging their complementary advantages.

In addition, experiments are conducted on four benchmark datasets, including the PaviaU, Houston, Honghu, and Hanchuan datasets, to validate the effectiveness of the proposed methods, achieving approximately 1.48%–2.51% increased OA, 1.34%–2.06% increased AA, and 1.37%–3.75% increased Kappa on these datasets, respectively. It highlights the proposed methods' superiority in HSI classification tasks.

II. Mamba-Based Methods in RS

This article introduces a Mamba network with FCL, which is applied to HSI classification tasks in RS. Therefore, its related work mainly involves the RSMamba methods and their developments in HSI classification tasks [9].

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According to different improvement methods, existing Mamba-based methods in RS can be divided into two categories: one enhances scan patterns to improve model computational efficiency, while the other enhances feature representation to improve model recognition performance [27]. For example, 3DSS-Mamba [21], Mamba-in-Mamba [28], and SpectralMamba [29] propose different enhanced scanning methods. Among them, 3DSS-Mamba [21] introduces a 3-D spectral–spatial selective scanning (3DSS) mechanism, which selectively scans 3-D hyperspectral tokens pixel by pixel. Mamba-in-Mamba [28] proposes a novel centralized Mamba-cross-scan mechanism that converts images into sequential data, achieving efficient scanning of RS images. SpectralMamba [29] uses a segmental scanning mechanism to convert approximately continuous spectra into compressed length sequences, effectively reducing computational load. Methods that improve Mamba feature representation include SS-Mamba [20], MambaHSI [24], S^2Mamba [23], Dual-Mamba [15], and LE-Mamba [30]. Among them, SS-Mamba [20], MambaHSI [24], and S^2Mamba [23] mainly focus on the joint spatial–spectral feature representation within the Mamba framework to enhance HSI classification performance. LE-Mamba [30] introduces a novel locally enhanced Mamba, while DualMamba [15] proposes a joint Mamba convolutional network to extract global and local spectral–spatial features, thereby enhancing HSI classification performance.

According to different frameworks, RSMamba can be divided into pure Mamba RS image interpretation models and RS image interpretation models that combine CNN or Transformer with Mamba. For example, the RSDehamba model [13] for dehazing RS satellite images, the RSM model [14] for dense prediction tasks in VHR RS images, and the new architecture RSMamba [9] for RS image classification are all based on the pure Mamba framework. DualMamba [15] combines a CNN module with the Mamba framework for HSI classification, effectively extracting global and local spectral–spatial features. RS3Mamba [16] constructs a RS semantic segmentation network by combining VSS blocks and CNN residual blocks. MamTrans [17] combines Transformer with the Mamba framework, effectively utilizing the Transformer's ability to capture spatial token relationships and Mamba's ability to extract implicit features along tokens. RTMamba [18] integrates convolutional blocks, Mamba's VSS blocks, and a semantic Transformer to build a real-time RS semantic segmentation model.

Overall, the Mamba-based methods in RS are rapidly advancing and have made significant research progress in HSI classification. In addition, combining CNN or Transformer to enhance the feature representation learning capability of Mamba has become a trend. However, in investigating Mamba methods used for HSI classification, it indicates that there are still some limitations in modeling the relationships between HSI's pixels and channel differences in existing work.

III. Local–Global Spectral Feature-Aware Learning for HSI Classification

The Mamba-based LGSFA-Net is constructed for HSI classification. It is detailed in this section, including the overall framework, Mamba block, IMamba block, and the ESSFF module.

A. Overall Framework of LGSFA-Net

The overall framework of the proposed LGSFA-Net is figured out in Fig. 2. It encompasses three stages, including the embedding, encoding, and classification stages. The embedding module consists of common convolutional layers, group normalization (GN), and SiLU activation function layers, enabling the extraction of embeddings for each pixel. Within the encoding stage, the Mamba block, IMamba block, and an ESSFF module are integrated together to form a comprehensive encoder. Specifically, upon receiving an input image, an embedding operation is conducted. Subsequently, the embeddings undergo processing by the Mamba block and IMamba block for feature extraction, respectively. The ESSFF module merges the features extracted by these two blocks. Following the aforementioned multilevel operations, the resulting features are directed to a classification (CLS) head for classification prediction, ultimately yielding the classification maps.

B. Mamba Block

The architecture of the Mamba block is based on the state-space model (SSM), which can extract the basic feature. The SSM can be expressed as the following linear ordinary differential equation. For input signals $x(t)$, the output $y(t)$ of SSM can be expressed as

$$h'(t) = Ah(t) + Bx(t) \tag{1}$$

$$y(t) = Ch(t) + Dx(t) \tag{2}$$

where $h'(t)$ represents the derivative of the state vector $h(t)$ with respect to time $dh(t)/dt$ and A, B, C, and D represent the state matrix, input matrix, output matrix, and feedforward matrix, respectively. Furthermore, it is extended to the 2-D image tasks with its convolutional form. In detail, the convolutional form's output SSM(x) can be expressed as

$$\mathrm{SSM}(x) = x * K \tag{3}$$

where $K = (\overline{CB}, \overline{CAB}, \ldots, \overline{CA}^k\overline{B}, \ldots,)$ and $\overline{A}, \overline{B}$, and $\overline{C}$ represent the discretization of A, B, and C, respectively.

The basic Mamba encoder is adapted to extract the input feature. For the input embedding H, there is the output y = Mamba(H) by the Mamba encoder. In detail, it can be expressed as follows:

$$y = \text{Linear Projection}(y1 \otimes y2) \tag{4}$$

$$y1 = \mathrm{SSM}\left(\delta\left(\text{Linear Projection}(H)\right)\right) \tag{5}$$

$$y2 = \delta\left(\text{Linear Projection}(H)\right) \tag{6}$$

where $\otimes$ represents the nonlinear projection (activation or multiplication operation), δ represents the SiLU activation function operation, and SSM() represents the SSM process. y_1 and y_2 represent intermediate feature outputs of the Mamba block (as shown in Fig. 3).

The proposed LGSFA-Net framework utilizes the visual Mamba encoder as the Mamba block, which is shown in Fig. 3. Based on the Mamba encoder above, the Mamba block can be

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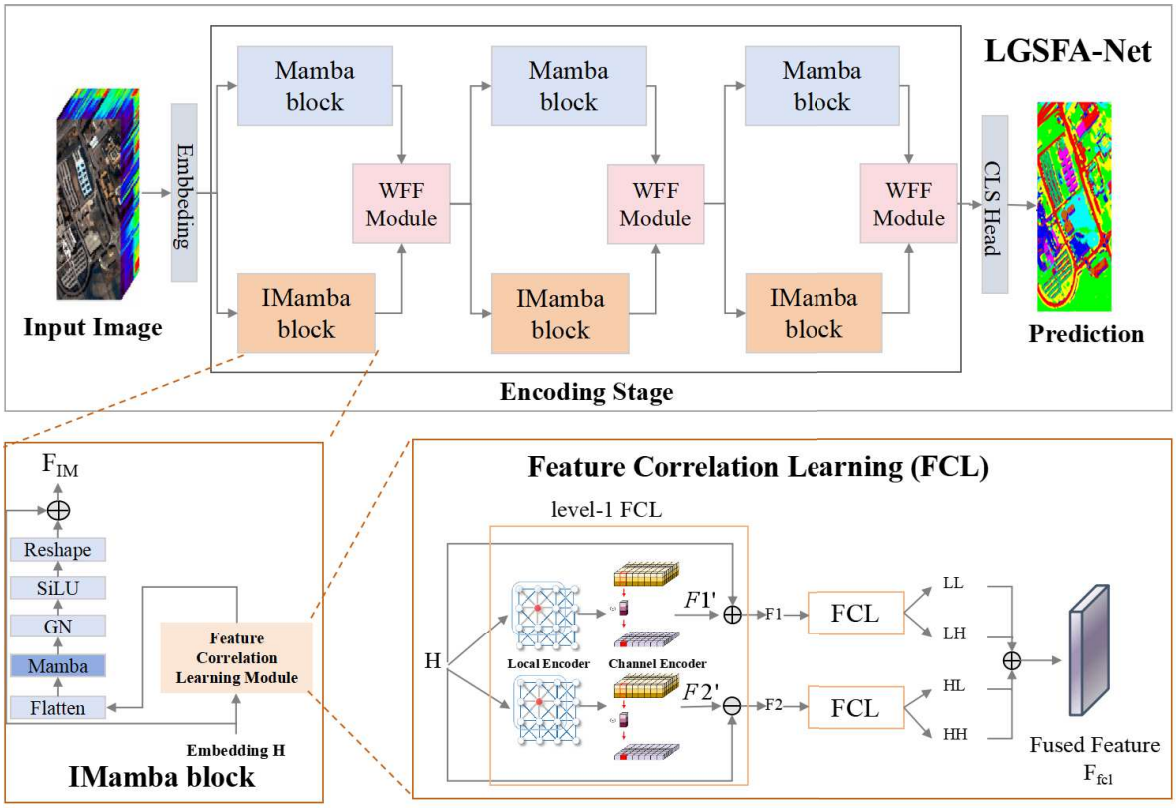


Fig. 2. Framework of the proposed LGSFA-Net framework. It mainly consists of the rmMamba block, the IMamba block, the ESSFF module, and the CLS head for final prediction.

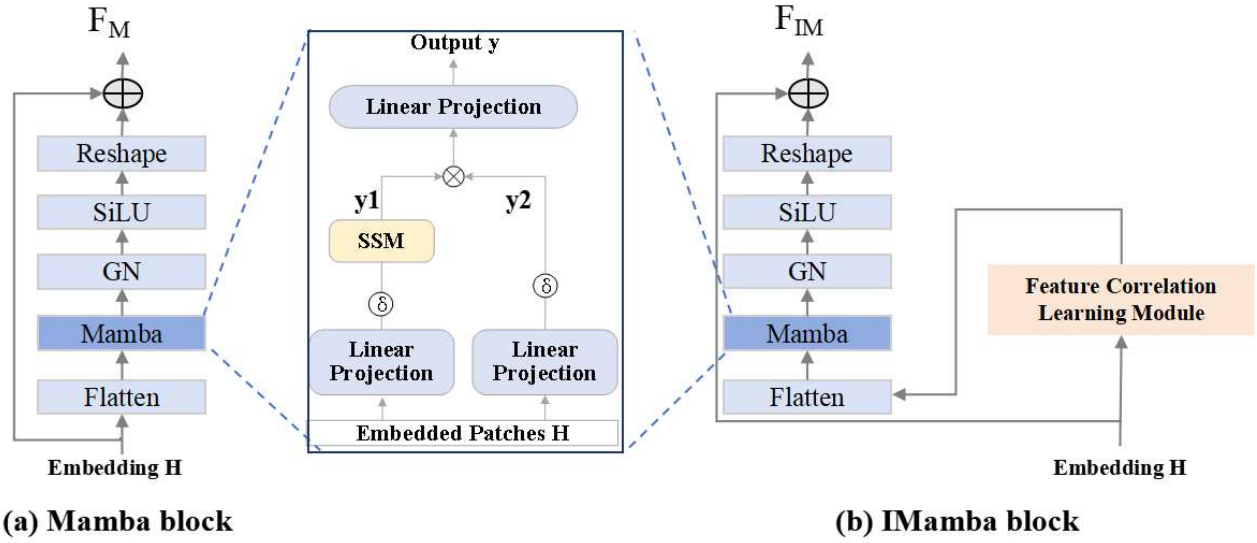


Fig. 3. Comparison of (a) Mamba block and (b) IMamba block. GN represents the group-normalization operation.

figured as Fig. 3(a). The output feature F_{M} can be formulated as follows. First, the embedding H can be obtained by the embedding operation

$$H = \mathrm{Embedding}\,(x)\,. \tag{7}$$

Also, then the embedding H is sent to the forward networks, which consist of flattening, Mamba encoding, GN, activation function, and the reshape operations. The feature F' can be expressed as

$$F' = \mathrm{SiLU}\,(\mathrm{GN}\,(\mathrm{Mamba}\,((\mathrm{Flatten}\,(H))))) \tag{8}$$

where SiLU represents the activation function. In addition, the Mamba block maintains the residual connection, thus its output F_{M} can be obtained as

$$F_{\mathrm{M}} = F' + H. \tag{9}$$

C. IMamba Block

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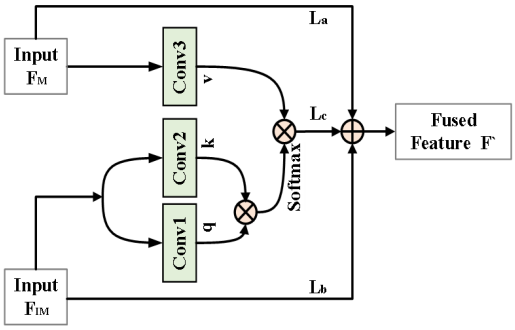


Fig. 4. ESSFF module.

The IMamba block with FCL is designed to consider the local and channel features based on the Mamba block. The architecture of the proposed IMamba block is shown in Fig. 3(b). Compared with the Mmaba block, the IMamba block introduces the FCL module. The output F_{IM} of IMamba block can be formulated as

$$F_{\mathrm{IM}} = \mathrm{SiLU}\left(\mathrm{GN}\left(\mathrm{Mamba}\left(\left(\mathrm{Flatten}\left(\mathrm{FCL}\left(H\right)\right)\right)\right)\right)\right). \tag{10}$$

In detail, the FCL module consists of the local encoder and the channel encoder. For the embedding H, the output of the FCL module can be formulated as follows:

$$F1 = H + F1' \tag{11}$$

$$F2 = H - F2' \tag{12}$$

where F is the encoding feature through relationship modeling. It can be expressed as

$$F1' = \mathrm{Channel\ Encoder1}\left(\mathrm{Local\ Encoder1}\left(H\right)\right) \tag{13}$$

$$F2' = \mathrm{Channel\ Encoder2}\left(\mathrm{Local\ Encoder2}\left(H\right)\right) \tag{14}$$

where Local Encoder() represents the convolutional operation encoding, which facilitates local feature extraction and aggregation. Channel Encoder() represents the channel relationship modeling. In detail, the channel correlation representation of the G = Local Encoder() can be formulated as

$$\mathrm{Channel\ Encoder}\left(G\right) = \mathrm{DWSConv}\left(G\right) + G \tag{15}$$

where DWSConv() represents the depthwise separable convolution. It is primarily used to reduce the number of parameters, increase nonlinearity, and achieve cross-channel information fusion. After depthwise convolution, channels are merged together again through 1×1 convolutions.

Based on the level-1 FCL operation, the FCL module utilizes level-2 FCL operation. Thus, the output of the FCL module can be expressed as

$$\begin{aligned} F_{\mathrm{fcl}} &= \mathrm{FCL}\left(F1\right) + \mathrm{FCL}\left(F2\right) \\ &= H1 + H2 + H3 + H4. \end{aligned} \tag{16}$$

D. Enhanced Spatial–Spectral Feature Fusion

Different sources of features often contain complementary information. To better capture the data features, we propose the utilization of a weighted fusion module to integrate the features derived from both the Mamba Block and IMamba Block. The ESSFF module is depicted in Fig. 4.

For the input F_{M} and input F_{IM}, the output F' of ESSFF module can be formulated as follows:

$$F' = L_a * F_{\mathrm{M}} + L_b * F_{\mathrm{IM}} + L_c * \mathrm{SA}\left(F_{\mathrm{M}}, F_{\mathrm{IM}}\right) \tag{17}$$

where L_a, L_b, and L_c represent the three learnable factors of different parts, and there is $L_a + L_b + L_c = 1$. The self-attention map can be figured out by setting the query (q), key (k), and value (v) in the self-attention module with $F_{\mathrm{M}}, F_{\mathrm{IM}}$ after convolution operations. It can be expressed as

$$\mathrm{SA}\left(F_{\mathrm{M}}, F_{\mathrm{IM}}\right) = \mathrm{Softmax}\left(F'_{\mathrm{IM}} * F''^{\mathrm{T}}_{\mathrm{IM}}\right) / \sqrt{d} * F'_{\mathrm{M}} \tag{18}$$

where SA() represents the self-attention operation, $F'_{\mathrm{IM}} = \mathrm{Conv1}(F_{\mathrm{IM}})$, $F''_{\mathrm{IM}} = \mathrm{Conv2}(F_{\mathrm{IM}})$, $F'_{\mathrm{M}} = \mathrm{Conv3}(F_{\mathrm{M}})$, and F_{M} and F_{IM} represent the feature extracted by the Mamba block and the feature extracted by the IMamba block, respectively. $F''^{\mathrm{T}}_{\mathrm{IM}}$ denotes the transpose of matrix F_{IM}. What is more, the self-attention is introduced to the calculation of the fused feature, which can better curve their pixel relationships with long-term independencies.

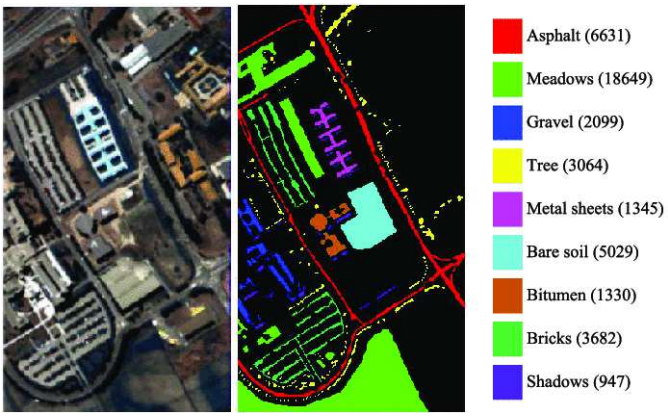


Fig. 5. Image and its ground truth of the PaviaU dataset.

IV. Experiments

This section covers the experimental setups, including the experimental dataset, settings, as well as experimental results and analysis. In addition, various ablation experiments are conducted to explore the impact of different factors on the proposed method. In addition, the theoretical analysis is validated through feature visualization.

A. Datasets

To further validate the effectiveness of the proposed LGSFA-Net, four benchmark datasets for HSI classification are chosen in experiments, including the PaviaU, Honghu, Hanchuan, and Houston datasets. The details of these four datasets are shown in Figs. 5–8. As for the PaviaU dataset, it is captured at the PaviaU in Italy. It contains high-resolution images in 103 spectral bands. It is divided into nine classes, with a total of 42 776 labeled samples. All the images have a resolution of 610×340 pixels, with each pixel representing an area of 1.3×1.3 m. These images depict various land cover types such as roads, buildings, grasslands, trees, and more. The example and its ground truth of the PaviaU dataset are shown in Fig. 5.

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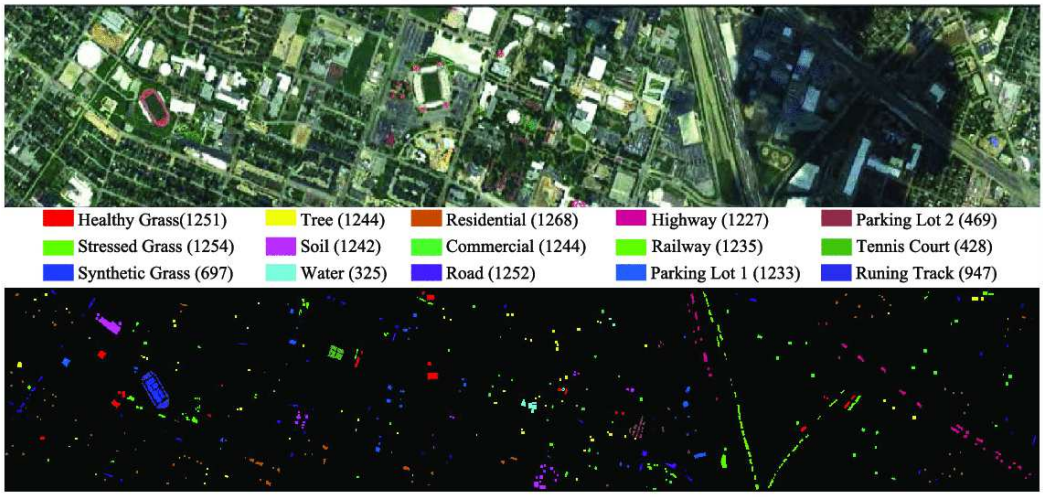


Fig. 6. Image and its ground truth of the Houston dataset.

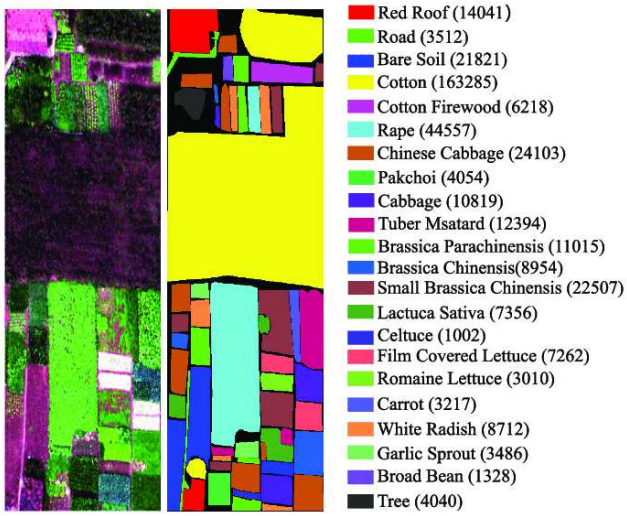


Fig. 7. Image and its ground truth of the Honghu dataset.

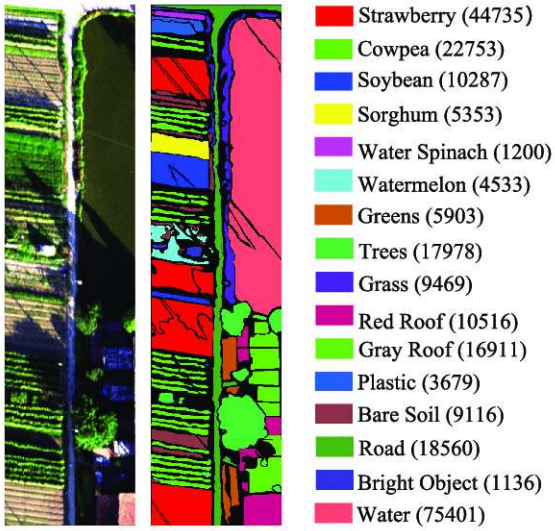


Fig. 8. Image and its ground truth of Hanchuan dataset.

As for the Houston dataset, it is provided by the Hyperspectral Image Analysis Laboratory and the National Science Foundation (NSF)-funded National Center for Airborne Laser Mapping (NCALM) at the University of Houston, USA. This dataset primarily captures the University of Houston campus and the surrounding urban area. It comprises approximately 400 HSIs covering 16 classes of land cover types, such as grasslands, trees, soil, water, and buildings. The dataset has a spatial resolution of 2.5 m, 144 spectral bands, and a spectral range from 380 to 1050 nm. The image and its ground truth of the Houston dataset are shown in Fig. 6.

The Hanchuan and Honghu datasets are high-spatial-resolution hyperspectral RS image datasets for fine-grained land cover classification constructed by the RSIDEA team of Wuhan University. The Honghu dataset encompasses diverse and complex agricultural scenes with numerous crop types, and the Hanchuan dataset involves agricultural scenes in regions combining urban and rural areas. The Honghu dataset consists of 22 classes (including 18 crop classes) with a spatial resolution of 0.043 m, an image size of 940 $\times$ 475 pixels, 270 bands, and a total of 386 693 samples. The Hanchuan dataset comprises 16 classes (including eight crop classes) with 274 bands, a resolution of 0.109 m, an image size of 1217 $\times$ 303 pixels, and a total of 257 530 samples. Specific class distributions and sample quantities for each class of the Honghu and Hanchuan datasets are shown in Figs. 7 and 8, respectively. In addition, both datasets have a spectral resolution of 6 nm.

In addition, in our experiments, we selected 30 and 10 samples of each class as the training and validation sets, respectively. Also, the rest of the samples are the testing set. All of them are normalized channelwise to $[0, 1]$ and then stretched to the range $[0, 255]$.

B. Settings and Implementation Details

All the experiments are conducted on the NVIDIA Titan V100 4 GPUs with the framework of PyTorch 1.7.1. The adaptive moment estimation (Adam) optimizer with beta of 0.9 is used for training. All the experiments are trained for 200 epochs. The learning rate is set at 0.0003. In addition, all

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TABLE I

EXPERIMENTAL RESULTS (%) ON PAVIAU DATASET. THE BEST IS GIVEN IN BOLD

Class	SVM	RF	DMSGer	GiGCN	FullyContNet	CLOLN	Spectralformer	GSC-ViT	MambaHSI	**Ours**
Asphalt	93.69±2.34	93.74±1.91	81.49±4.78	90.36±1.73	90.98±2.47	**98.27±1.00**	92.60±2.29	97.10±1.64	94.35±1.73	95.75±1.49
Meadows	88.56±2.41	87.29±1.93	87.60±3.03	98.86±0.78	89.37±1.81	**98.98±0.76**	93.58±2.14	97.68±1.32	96.43±1.92	97.12±1.25
Gravel	50.42±5.48	48.47±4.84	97.15±1.24	82.64±3.09	91.37±5.61	**97.71±1.51**	47.98±2.13	60.29±3.96	93.39±3.55	**98.69±0.65**
Trees	55.88±5.67	55.21±4.85	87.68±2.85	81.34±10.11	93.50±6.74	**97.80±1.46**	92.28±2.23	89.75±3.64	89.75±3.64	96.88±0.72
Metal Sheets	95.67±1.04	95.21±1.02	98.90±0.22	98.77±0.19	98.48±0.45	98.97±0.27	96.59±1.14	97.80±0.31	**99.75±0.12**	97.32±1.43
Bare Soil	49.11±8.63	44.29±5.43	**99.66±0.60**	**99.66±0.60**	85.43±8.76	85.43±8.76	63.92±2.41	48.10±4.58	85.43±8.76	**99.88±0.11**
Bitumen	41.46±3.09	48.42±4.67	**99.36±0.88**	99.28±0.61	79.28±8.56	87.87±4.63	92.10±2.55	94.65±1.07	98.79±0.84	98.76±0.57
Bricks	73.17±4.64	73.68±4.58	86.96±1.74	87.43±1.54	89.67±4.64	**96.73±2.90**	79.60±3.61	95.27±0.73	94.40±2.93	**97.66±0.79**
Shadows	**99.91±0.08**	99.81±0.11	96.89±0.82	97.12±0.82	96.73±9.02	97.12±2.03	88.78±5.92	98.71±1.10	99.36±0.62	99.12±0.32
OA(%)	71.31±3.15	70.68±2.61	91.31±0.97	91.31±0.97	91.31±0.97	93.53±1.59	82.45±1.64	91.31±0.97	95.74±0.90	**97.22±0.41**
AA(%)	71.98±1.39	71.14±1.61	91.17±0.97	91.17±0.97	91.17±0.97	93.53±1.59	77.40±4.41	91.17±0.97	95.86±1.11	**97.92±0.38**
Kappa(%)	63.96±3.19	63.13±2.81	86.49±1.54	92.67±1.26	93.12±5.19	93.38±1.84	77.40±4.41	90.38±2.95	95.00±2.24	**98.75±0.85**

TABLE II

EXPERIMENTAL RESULTS (%) ON HOUSTON DATASET. THE BEST IS GIVEN IN BOLD

Class	**SVM**	**RF**	**DMSGer**	**GiGCN**	**FullyContNet**	**CLOLN**	**Spectralformer**	**GSC-ViT**	**MambaHSI**	**Ours**
Healthy Grass	93.68±5.88	93.00±0.45	91.27±4.59	83.21±5.03	92.69±7.57	88.24±9.00	91.29±5.38	90.16±6.48	**95.20±4.66**	94.10±1.32
Stressed Grass	85.92±6.74	91.94±1.40	92.24±4.73	82.83±7.41	95.06±1.86	95.77±1.49	93.32±3.99	95.77±2.49	**98.29±1.26**	98.00±1.54
Synthetic Grass	98.83±1.38	97.82±3.65	99.62±0.39	98.83±1.66	99.21±0.74	95.21±4.90	94.68±3.25	**99.83±0.26**	99.74±0.46	**99.94±0.06**
Tree	**98.89±0.50**	**98.89±0.50**	83.67±4.86	91.34±4.62	97.47±4.48	93.48±3.64	97.47±4.48	98.83±1.66	93.32±7.04	97.50±0.65
Soil	90.87±2.41	87.75±4.04	**98.76±1.79**	93.14±4.62	97.07±6.99	97.07±6.99	98.18±1.72	97.32±3.69	95.44±0.76	**99.44±0.56**
Water	87.86±14.30	87.30±6.08	99.79±0.63	99.09±4.78	97.58±1.78	89.54±8.72	87.21±1.86	99.70±0.03	99.70±0.03	**99.97±0.03**
Residential	73.99±4.64	74.97±4.97	67.40±5.02	91.02±6.25	76.74±4.92	93.04±3.94	84.41±7.91	88.35±6.34	92.22±2.69	**98.00±0.35**
Commercial	79.40±11.34	83.26±3.92	67.40±5.02	92.50±1.95	85.40±9.92	**93.04±3.94**	84.41±7.91	88.64±3.43	91.94±2.62	86.70±2.54
Road	65.11±4.19	64.21±4.64	84.64±2.18	92.79±1.94	91.88±9.34	**94.21±3.54**	91.32±7.39	91.58±8.34	94.16±3.56	90.80±2.02
Highway	70.50±4.76	73.48±6.10	98.16±1.89	94.48±2.56	90.51±6.44	92.08±4.27	89.84±5.28	90.48±5.28	95.44±2.76	**99.36±0.64**
Railway	70.15±2.45	66.13±3.24	94.16±1.34	96.52±3.78	93.34±4.93	94.44±4.40	91.32±5.28	89.84±5.28	94.42±3.79	**99.50±0.03**
Parking Lot 1	64.23±4.99	58.37±5.03	92.52±3.00	92.45±3.63	89.28±3.32	**92.98±3.32**	88.01±7.56	91.49±4.10	92.95±3.30	92.00±1.23
Parking Lot 2	43.46±10.62	31.71±4.80	96.95±1.24	**100.00±0.00**	92.98±3.32	**100.00±0.00**	88.01±7.56	92.98±3.28	**100.00±0.00**	97.00±2.28
Tennis Court	88.44±5.01	80.89±5.68	99.47±0.10	99.14±1.89	97.07±0.74	97.67±0.62	96.92±7.50	94.92±3.40	97.67±0.62	**100.00±0.00**
Running Track	**99.16±0.46**	97.08±1.84	97.00±1.34	99.14±1.89	98.47±3.70	98.47±3.70	96.18±2.66	97.08±1.84	99.14±1.89	97.70±0.42
OA(%)	79.63±1.04	78.95±0.90	90.36±1.21	91.31±1.26	90.64±1.44	92.57±1.12	88.92±1.56	91.31±1.26	94.66±0.97	**96.20±0.32**
AA(%)	80.70±1.04	79.24±0.83	92.17±1.07	91.85±1.91	91.71±1.35	92.17±1.07	87.92±1.58	92.17±1.07	95.34±0.78	**96.75±0.58**
Kappa(%)	77.97±1.12	77.25±0.98	89.58±1.39	87.16±2.44	90.64±1.68	90.64±1.68	86.92±1.80	91.18±1.14	94.21±1.93	**96.40±1.32**

experiments were conducted three times (with seed = 1,2,3), and the average± standard deviation values were selected for the final performance evaluation.

In experiments, three commonly used evaluation metrics are adopted, including the OA, AA, and the Kappa coefficient (k). In addition, the CrossEntropyLoss is selected as the loss function. In addition, the CLS head adopts the simple classification head with two layers of 2-D convolution operation, with GN and SiLU operations inserted.

Empirically designed, the local encoder process utilizes 1×4 and 4×1 convolutional kernels instead of standard 4×4 kernels, significantly reducing the number of parameters and computational load while achieving a similarly acceptable receptive field, with the number of channels consistent with the input. The channel encoder's depthwise separable convolution employs fixed parameters (kernel size = 3×3, padding = 1, and stride = 1), and the number of groups equals the input channels (achieving depthwise separation). In addition, L_a, L_b, and L_c are initialized with equal weights of one-third each.

C. Contrasting Methods

In this study, experimental comparisons of HSI classification methods are conducted. First, we investigated machine learning-based methods, including support vector machine (SVM) [31] and random forest (RF) [32], and their utility in handling HSI classification tasks. Second, we studied graph convolutional network (GCN)-based methods, such as the dynamic multiscale GCN classifier (DMSGer) [33] and the graph-integrated GCN (GiGCN) [34], which enhance classification accuracy by capturing pixel-level and region-level features. In addition, we explored CNN-based methods, such as FullyContNet [35] and CLOLN [36], which utilize different feature extractors and network structures to enhance classification context and reduce computational complexity. Finally, we discussed Transformer-based methods, including Spectralformer [37] and groupwise separable convolution Vision Transformer (GSC-ViT) [38], which improve HSI classification performance by extracting local spectral information and global spectral–spatial information. Of course, the MambaHSI [24] is selected as the contrasting methods too. It is the first model to explore the Mamba's potential in HSI classification.

D. Experimental Results and Analysis

1) Performance on PaviaU Dataset: As shown in Table I, the proposed method in this article demonstrated OA, AA, and Kappa performances of 97.22%, 97.92%, and 98.75%, respectively, on the PaviaU dataset. Compared with other contrastive methods, the proposed approach exhibited superior performance. Specifically, within machine learning methods, the SVM method showed OA, AA, and Kappa performances of 71.31%, 71.98%, and 63.96%, respectively. Among GCN-based methods, GiGCN achieved OA and Kappa performance gains of approximately 4.92% and 6.18% over the

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TABLE III

EXPERIMENTAL RESULTS (%) ON HONGHU DATASET. THE BEST IS GIVEN IN BOLD

Class	SVM	RF	DMSGer	GiGCN	FullyContNet	CLOLN	Spectralformer	GSC-ViT	MambaHSI	Ours
Red Roof	94.30±1.90	84.35±5.74	92.96±4.02	90.93±2.64	96.78±1.57	95.36±5.56	96.66±1.63	**98.16±1.08**	96.47±1.50	95.89±1.12
Road	53.77±4.54	52.01±5.17	93.34±4.05	51.88±7.77	95.63±2.54	78.60 ±6.19	62.84±4.24	73.78±9.09	95.43±2.33	**98.91±0.56**
Bare Soil	90.7±3.65	90.92±2.7	92.15±3.46	78.8±2.87	92.23±1.51	**98.22±1.32**	95.6±2.81	97.21±1.81	94.66±1.05	91.39±2.23
Cotton	97.42±0.34	97.09±0.51	60.26±10.53	99.37±0.32	93.59±2.92	99.48±0.37	99.42±0.30	**99.57±0.14**	96.96±1.79	98.73±0.88
Cotton Firewood	17.61±2.78	14.17±2.64	**99.30±0.79**	62.41±8.43	96.03±2.60	41.33±18.92	42.14±16.25	37.03±9.33	96.20±1.77	99.14±0.11
Rape	90.59±1.93	84.04±1.60	84.36±4.27	89.61±3.71	91.61±2.28	**96.09±1.67**	95.73±1.02	95.72±1.83	93.90±2.47	95.75±1.55
Chinese Cabbage	78.6±3.21	72.03±4.93	69.44±3.44	82.91±3.07	82.52±3.43	92.72±1.96	90.05±2.26	90.38±3.66	86.60±3.30	**95.75±2.11**
Pakchoi	13.56±2.31	9.13±2.07	**98.36±2.87**	75.44±8.66	96.81±2.03	46.44±6.02	35.59±5.04	32.12±4.83	96.08±2.90	89.73±3.65
Cabbage	96.57±1.71	90.6±3.89	95.14±1.73	85.39±3.04	96.62±1.20	97.04±2.05	**97.89±1.07**	97.06±2.73	94.78±1.69	94.05±2.22
Tuber Msatard	50.25±7.08	33.29±6.19	87.22±4.16	92.42±2.80	90.08±2.81	84.88±3.47	78.81±4.58	79.23±7.53	90.45±2.72	**98.45±0.65**
Brassica Parachinensis	31.7±5.48	25.71±4.48	82.08±7.09	75.17±4.5	86.27±3.66	72.03±8.35	69.29±7.09	72.41±8.95	91.96±2.87	**96.14±1.54**
Brassica Chinensis	42.58±4.37	40.77±3.36	91.91±2.46	82.52±5.36	88.62±4.09	66.63±8.82	58.68±5.96	65.61±8.66	87.40±5.22	**94.44±2.67**
Small Brassica Chinensis	59.05±4.52	54.24±3.33	73.10±6.40	89.28±2.70	89.16±2.24	82.67±5.83	76.20±6.07	77.49±8.47	89.12±3.12	**96.88±0.98**
Lactuca Sativa	67.70±6.77	57.66±7.96	**94.87±2.43**	68.03±5.44	93.88±1.93	87.59±7.03	83.89±6.15	86.26±11.93	94.01±1.65	90.84±2.14
Celtuce	8.86±4.25	3.78±1.16	**99.85±0.28**	21.65±2.65	98.42±1.16	75.88±12.93	61.24±13.37	63.62±20.25	98.84±1.17	94.12±1.22
Film Covered Lettuce	87.85±2.55	80.73±7.57	96.89±1.26	71.70±6.79	88.96±4.28	97.39±1.73	95.23±2.99	97.42±1.66	96.83±1.60	**99.35±0.21**
Romaine Lettuce	55.60±4.87	54.14±6.89	**99.23±1.01**	54.94±4.04	95.10±4.52	80.09±9.70	68.68±8.79	84.92±6.09	98.60±1.49	98.01±1.21
Carrot	27.46±3.59	18.32±1.20	**98.44±2.49**	67.65±8.17	97.22±1.43	65.55±15.89	60.99±7.01	67.44±10.50	96.72±1.82	96.63±2.44
White Radish	65.15±5.12	47.85±5.57	88.34±4.02	72.71±8.86	89.53±3.28	82.53±7.23	82.77±4.78	85.10±8.11	91.30 ±3.07	**93.50±1.20**
Garlic Sprout	37.85±5.53	17.29±4.7	99.02±0.85	66.48±6.47	98.34±1.10	81.38±6.60	71.18±8.29	71.64±10.46	**99.18±0.42**	97.71±0.78
Broad Bean	9.35±0.92	9.68±0.83	99.96±0.12	65.20±14.58	99.95±0.12	30.19±12.06	28.91±8.18	25.92±7.91	99.91±0.18	**100.00±0.00**
Tree	21.53±1.27	20.00±1.27	**99.88±0.24**	85.35±8.84	99.11±1.00	55.02±10.89	58.01±15.28	46.26±11.15	99.54±0.51	96.58±1.32
OA(%)	68.77±1.27	59.85±2.13	75.43±4.34	87.22±0.60	92.19±1.35	87.93±2.11	86.04±2.83	86.18±1.86	94.58±1.01	**96.25±1.59**
AA(%)	54.50±0.41	48.08±1.05	90.73±0.92	74.08±1.50	93.47±0.48	77.60±2.25	73.17±2.45	74.74±1.64	94.77±0.61	**96.11±0.52**
Kappa(%)	62.71±1.28	53.22±1.95	71.59±4.32	83.95±0.73	93.86±1.54	83.95±2.49	82.72±3.22	82.93±2.16	94.00±1.88	**96.67±0.74**

TABLE IV

EXPERIMENTAL RESULTS (%) ON HANCHUAN DATASET. THE BEST IS GIVEN IN BOLD

Class	SVM	RF	DMSGer	GiGCN	FullyContNet	CLOLN	Spectralformer	GSC-ViT	MambaHSI	Ours
Strawberry	92.03±1.24	87.87±1.73	74.51±5.53	89.34±2.23	78.59±12.03	**95.97±1.87**	95.13±1.25	95.26±1.5	89.72±5.75	95.05±1.16
Cowpea	69.50±7.79	57.02±4.42	62.51±8.18	81.64±3.38	84.21±4.34	**92.48±2.56**	87.27±3.37	91.22±3.05	81.37±5.69	87.44±0.93
Soybean	47.41±4.94	45.31±2.24	94.63±2.80	81.06±6.56	79.85±9.55	80.03±9.18	70.36±6.15	78.37±7.47	94.88±3.73	**97.57±1.60**
Sorghum	74.49±9.57	51.50±2.59	99.51±0.50	67.09±4.82	84.68±13.06	91.12±7.72	88.62±6.67	85.17±12.38	98.46±1.35	**99.10±0.47**
Water Spinach	9.72±0.77	9.25±0.78	99.18±0.92	34.98±2.76	68.38±27.44	56.82±12.89	42.01±11.48	46.57±13.03	99.54±0.85	**99.91±0.12**
Watermelon	15.28±2.00	11.66±1.03	**94.18±2.43**	53.81±9.81	9.39±15.90	38.07±8.70	34.68±6.40	35.97±7.01	76.99±4.95	72.45±6.10
Greens	36.69±2.30	45.58±3.46	**98.72±1.26**	80.84±8.87	92.94±4.47	68.87±17.46	67.69±10.29	66.90±15.67	92.61±4.63	92.65±4.48
Trees	65.71±5.51	66.22±4.84	84.57±4.09	89.27±4.80	**91.24±1.96**	81.52±7.87	83.17±4.16	82.39±5.23	70.73±7.10	75.23±1.89
Grass	47.93±7.71	40.87±7.40	72.25±5.64	65.14±6.79	70.48±8.43	68.59±6.61	63.00±4.53	60.64±17.62	**89.01±3.62**	85.33±2.72
Red Roof	74.50±11.85	59.59±6.64	87.81±3.96	93.03±4.21	85.23±6.83	90.79±14.71	88.76±12.99	93.22±4.31	94.44±2.18	**97.87±0.82**
Gray Roof	35.90±1.90	54.20±9.34	**96.13±1.66**	93.86±5.06	92.58±4.29	87.86±4.63	77.41±13.19	82.13±7.96	89.56±4.91	92.35±3.31
Plastic	13.49±2.57	17.50±2.66	**99.09±1.04**	89.22±7.45	39.55±27.62	63.48±14.8	51.40±6.89	51.92±6.01	88.83±6.91	92.72±2.46
Bare Soil	27.36±3.66	33.89±4.49	**82.67±3.97**	66.66±6.60	82.37±2.46	50.21±6.71	47.62±9.17	53.28±10.88	78.69±3.73	80.01±4.16
Road	80.55±8.56	76.20±6.53	75.85±3.16	91.25±2.43	88.31±4.97	90.78±2.74	88.92±3.08	90.64±3.07	89.68±3.03	**94.04±0.92**
Bright Object	33.15±10.97	26.80±10.82	**98.36±2.21**	46.57±17.42	97.30±2.18	62.42±10.67	52.51±8.59	70.05±12.35	94.42±2.41	96.78±0.87
Water	**99.95±0.06**	98.82±1.33	92.23±5.31	99.75±0.14	93.64±4.75	98.97±1.02	99.47±0.33	99.77±0.23	98.28±1.62	98.55±0.68
OA(%)	60.98±1.35	64.09±1.78	84.39±1.83	86.78±1.43	84.88±3.02	85.14±2.93	82.99±2.31	83.14±3.96	90.21±1.67	**92.73±0.34**
AA(%)	51.48±1.46	48.89±1.15	88.26±0.89	76.47±1.62	77.42±3.61	76.13±3.26	71.13±2.56	73.97±2.06	89.20±1.36	**91.06±0.66**
Kappa(%)	56.00±1.49	59.05±1.86	82.01±2.02	84.64±1.62	75.80±4.7	82.75±3.31	80.29±2.61	80.57±4.33	89.07±2.31	**90.44±2.74**

TABLE V

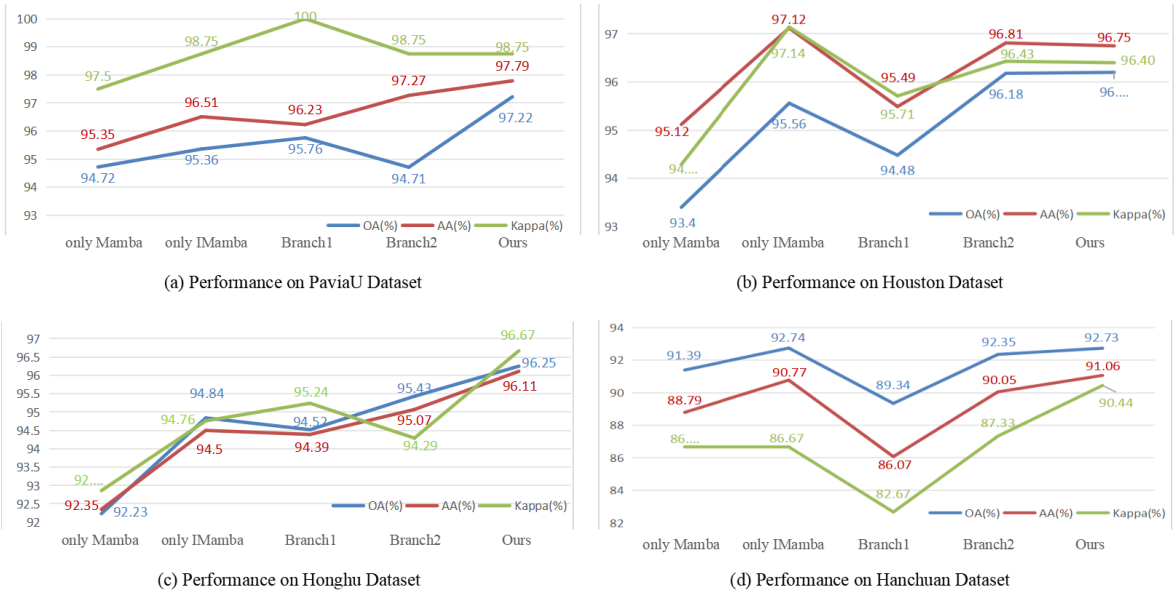
COMPARISON PERFORMANCE OF THREE DIFFERENT FACTOR SETTINGS OF ESSFF MODULE. THE BEST IS GIVEN IN BOLD

Settings				PaviaU			Houston			Honghu			Hanchuan		
Modes	La	Lb	Lc	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
SA(F_M,F_{IM}) Only	0	0	1	70.45±1.40	75.81±0.57	69.17±7.24	74.48±2.44	77.10±3.24	69.05±3.88	81.06±2.5	84.08±0.75	83.02±3.84	65.98±2.61	56.45±4.46	53.56±4.94
F_{IM} Only	0	1	0	95.24±1.55	96.91±0.32	96.67±2.12	96.45±0.16	**96.99±0.22**	96.19±1.35	94.30±0.52	93.93±0.68	93.65±1.96	92.74±0.32	90.73±0.25	89.78±1.13
F_M Only	1	0	0	95.90±0.66	95.98±0.36	95.83±3.28	95.00±0.45	95.84±0.47	94.76±1.47	92.60±2.48	93.28±1.14	93.81±2.16	89.83±0.83	88.43±1.27	88.44±3.28
Average Fusion	1/3	1/3	1/3	97.16±0.31	97.93±0.38	97.92±2.12	96.31±0.20	96.84±0.19	95.95±1.21	95.84±0.33	95.76±0.32	96.03±0.90	92.94±0.34	91.45±0.58	90.22±2.45
F_M+F_{IM}	0.5	0.5	0	97.27±0.28	**98.04±0.25**	97.92±2.12	96.25±0.73	96.82±0.68	96.19±0.89	95.85±0.31	95.77±0.31	95.56±1.25	92.91±0.49	91.56±0.40	**90.89±2.57**
F_M-Preferred	0.6	0.2	0.2	97.43±0.55	97.98±0.56	97.08±3.28	96.01±0.65	96.59±0.61	95.95±1.78	95.92±0.62	95.75±0.28	95.87±1.12	92.60±0.36	91.19±0.76	90.22±4.53
F_{IM}-Preferred	0.2	0.6	0.2	97.02±1.11	97.80±0.16	97.50±2.70	**96.33±0.32**	96.90±0.35	96.43±1.54	95.02±0.52	95.13±0.33	94.29±1.17	92.55±0.48	91.01±0.39	90.22±3.00
SA(F_M,F_{IM})-Preferred	0.2	0.2	0.6	**97.44±0.34**	97.86±0.27	97.50±2.70	96.13±0.35	96.74±0.25	**96.43±0.58**	95.34±0.88	95.40±0.58	95.87±1.12	92.6±0.93	90.84±1.40	90.22±2.45
Ours(Learnable)	Le	Le	Le	97.22±0.41	97.92±0.38	**98.75±0.85**	96.20±0.32	96.75±0.58	96.40±1.32	**96.25±1.59**	**96.11±0.52**	**96.67±0.74**	**93.21±1.52**	**91.94±1.22**	90.44±2.74

DMSGer method. In CNN-based approaches, the CLOLN model significantly outperformed the FullyContNet model. In Transformer-based methods, GSC-ViT outperformed the SpectralFormer method by 1.55%, 11.96%, and 12.98% in terms of OA, AA, and Kappa performance. MambaHSI attempted HSI classification tasks with Mamba, achieving OA, AA, and Kappa performances of 95.74%, 95.86%, and 95% on the PaviaU dataset, showcasing a certain performance advantage over comparative methods. Compared with the MambaHSI, the proposed method still demonstrated OA, AA, and Kappa performance gains of 1.48%, 2.06%, and 3.75%. Furthermore, it achieved optimal performance across three categories, including Gravel, Bare Soil, and Bricks.

2) Performance on Houston Dataset: As shown in Table II, the proposed LGSFA-Net demonstrated OA, AA, and Kappa

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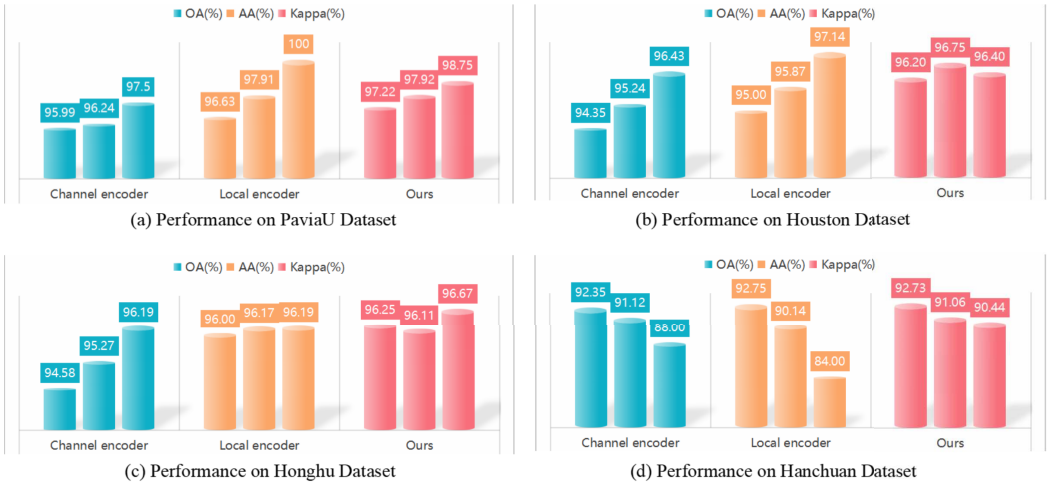
(a) Performance on PaviaU Dataset

(b) Performance on Houston Dataset

(c) Performance on Honghu Dataset

(d) Performance on Hanchuan Dataset

Fig. 9. Compared with the performance of four frameworks on (a) PaviaU, (b) Houston, (c) Honghu, and (d) Hanchuan datasets. It includes the only Mamba-based, only IMamba-based, Mamba + ESSFF-based, IMamba + ESSFF-based, and the LGSFA-Net-based modes.



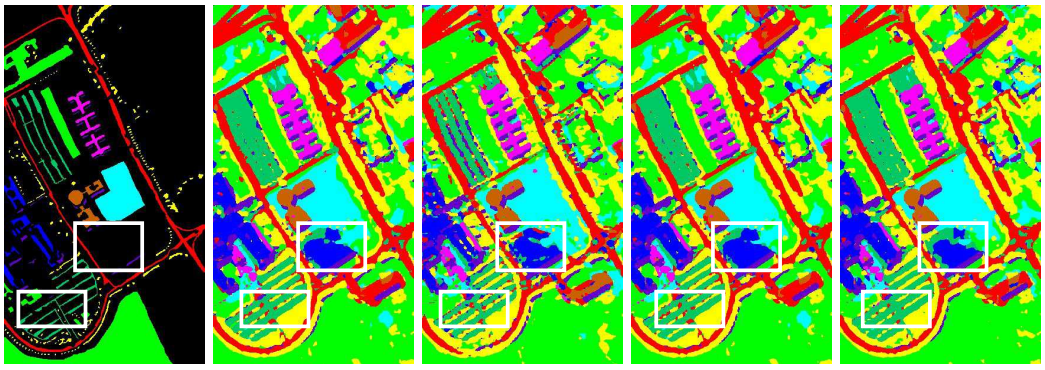
(a) Performance on PaviaU Dataset

(b) Performance on Houston Dataset

(c) Performance on Honghu Dataset

(d) Performance on Hanchuan Dataset

Fig. 10. Compared performance of LGSFA-Net framework with local encoder and channel encoder on (a) PaviaU, (b) Houston, (c) Honghu, and (d) Hanchuan datasets.



(a) (b) (c) (d) (e)

Fig. 11. Visualization of experimental results on PaviaU dataset, including the feature visualization of (a) ground truth, (b) MambaHSI, (c) Branch1, (d) Branch2, and (e) LGSFA-Net.

performances of 96.20%, 96.75%, and 96.40%, respectively, on the Houston dataset. Compared with other contrastive methods, the proposed approach exhibited superior performance. Specifically, within machine learning methods, the SVM method and the RF method showed comparable performance on the Houston dataset. Among GCN-based

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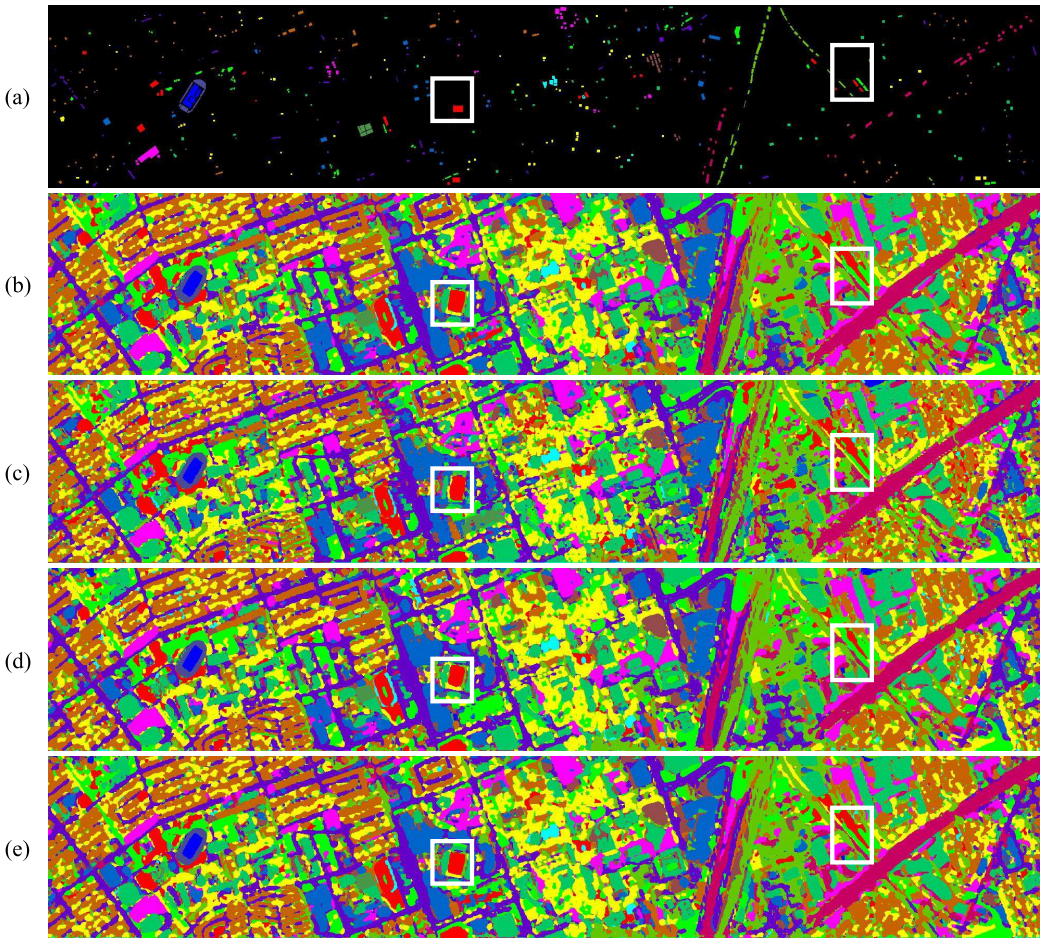


Fig. 12. Visualization of experimental results on Houston dataset, including the feature visualization of (a) ground truth, (b) MambaHSI, (c) Branch1, (d) Branch2, and (e) LGSFA-Net.

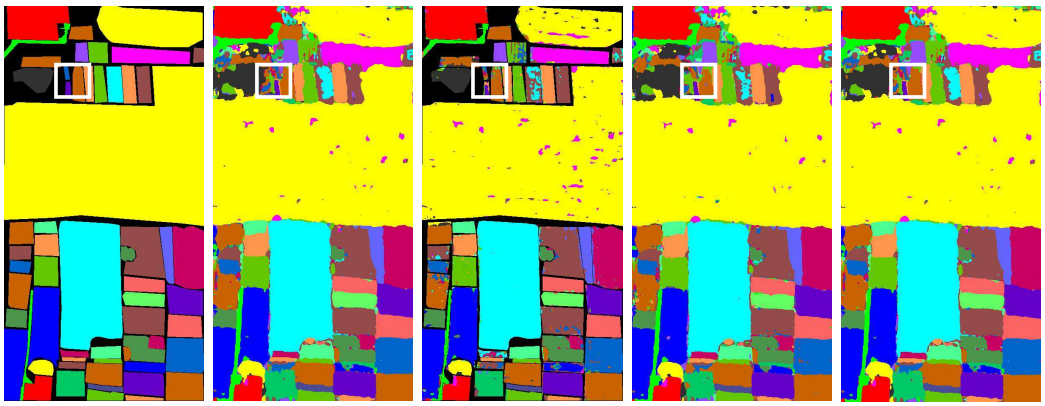


Fig. 13. Visualization of experimental results on Honghu dataset, including the feature visualization of (a) ground truth, (b) MambaHSI, (c) Branch1, (d) Branch2, and (e) LGSFA-Net.

methods, DMSGer achieved OA, AA, and Kappa performance gains of approximately 2.05%, 2.32%, and 2.42% over the GiGCS method. In CNN-based approaches, the CLOLN model and FullyContNet showed competitive performance on Houston. Within Transformer-based methods, GSC-ViT outperformed the SpectralFormer method by 2.72%, 3.08%, and 2.94% in terms of OA, AA, and Kappa performance. MambaHSI achieves OA, AA, and Kappa performance of 94.66%, 95.34%, and 94.21% on the Houston dataset, showcasing an absolute performance advantage over comparative methods. Compared with the MambaHSI approach, the proposed method still demonstrated OA, AA, and Kappa performance gains of 1.54%, 1.41%, and 2.19%. Furthermore, it achieved optimal performance across seven categories, including Synthetic Grass, Soil, Water, Highway, Residential, Railway, and Tennis Court.

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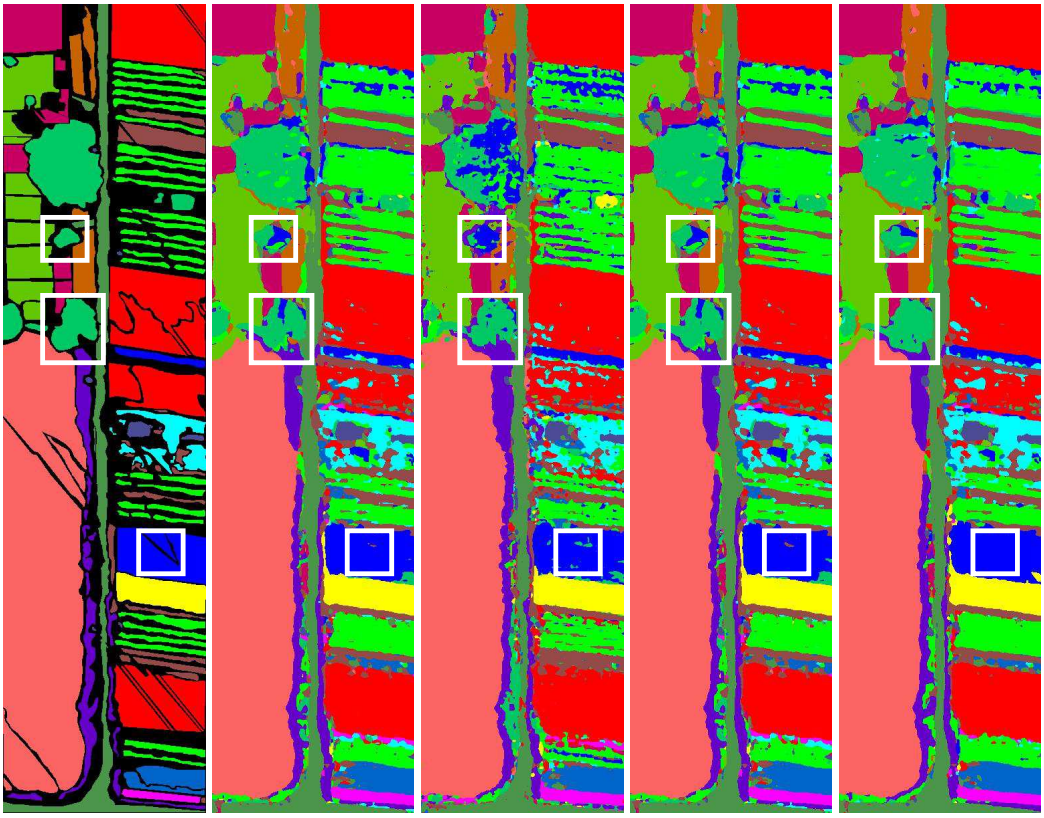


Fig. 14. Visualization of experimental results on Hanchuan dataset, including the feature visualization of (a) ground truth, (b) MambaHSI, (c) Branch1, (d) Branch2, and (e) LGSFA-Net.

3) Performance on Honghu Dataset: As shown in Table III, the proposed LGSFA-Net demonstrated OA, AA, and Kappa performances of 96.25%, 96.11%, and 96.67%, respectively, on the Honghu dataset. Compared with other contrastive methods, our method achieved the best performance on this dataset. Similar to the Hanchuan dataset, most comparative algorithms also showed relatively low classification accuracy on the Honghu dataset. In contrast, FullyContNet and MambaHSI achieved significantly higher classification performance on the Honghu dataset compared with other contrastive methods, with MambaHSI outperforming FullyContNet. The proposed method in this article achieved OA, AA, and Kappa performance gains of approximately 1.67%, 1.34%, and 2.67% higher than MambaHSI on the Honghu dataset. In addition, the proposed method achieved optimal performance across eight categories, including Road, Chinese Cabbage, White Radish, and Tree.

4) Performance on Hanchuan Dataset: As shown in Table IV, the proposed LGSFA-Net demonstrated OA, AA, and Kappa performances of 92.73%, 91.06%, and 90.44%, respectively, on the Hanchuan dataset. Compared with other contrastive methods, the proposed approach exhibited superior performance in terms of OA and AA metrics. Unlike the PaviaU and Houston datasets, the comparative algorithms showed relatively low classification accuracy on the Hanchuan dataset. MambaHSI achieved classification performance significantly higher than other comparative methods. The proposed LGSFA-Net achieved OA, AA, and Kappa performance gains of approximately 2.52%, 1.86%, and 1.37% higher than MambaHSI on the Hanchuan dataset. In addition, it achieved optimal performance across five categories, including Soybean, Sorghum, Water Spinach, Red Roof, and Road.

In summary, the outstanding performance of the proposed LGSFA-Net method on multiple benchmarking datasets indicates its higher generality and robustness compared with the contrasting methods, suggesting that it is more suitable for a wider range of HSI classification tasks.

5) Ablations: Regarding the ablation experiments, we conducted experiments on the framework structure, the presence or absence of the feature fusion module, as well as ablation experiments on the CNN and channel encoder involved in FCL.

a) Ablation on the framework: To evaluate the effect of the proposed module in the LGSFA-Net framework, the ablation experiments are done on the proposed module. There are four structures considered to compare with that only based on the Mamba block, including Mamba block with FCL-based (IMamba), Branch1 (Mamba block + ESSFF-based), Branch2 (IMamba block + ESSFF-based), and the proposed LGSFA-Net framework. The experimental results are shown in Fig. 9. It shows that the framework only based on the Mamba block with FCL (IMamba) performs better than that of only the Mamba block based on all four benchmark datasets, and the improved performance is obvious. In addition, Branch2's performance is superior to that of Branch1's. Furthermore, it shows that Branch2, based on IMamba and ESSFF

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TABLE VI
COMPARE AVERAGE PERFORMANCE (%) OF VISUALIZATION METHODS

PaviaU\Methods	MambaHSI	Branch1	Branch2	Ours
Asphalt	94.35	90.14	**96.81**	95.75
Meadows	96.43	**97.16**	91.40	97.12
Gravel	93.39	95.24	**99.42**	98.69
Trees	89.75	90.77	95.77	**96.88**
Metal Sheets	99.75	**100.00**	**100.00**	97.32
Bare Soil	85.43	99.22	97.47	**99.88**
Bitumen	98.79	98.91	**99.22**	98.76
Bricks	94.40	94.87	95.80	**97.66**
Shadows	99.36	**99.78**	99.56	99.12
OA(%)	95.74	95.76	94.71	**97.22**
AA(%)	95.86	96.23	97.27	**97.92**
Kappa(%)	95.00	**100.00**	98.75	98.75
Houston\Methods	MambaHSI	Branch1	Branch2	Ours
Healthy Grass	**95.20**	94.55	91.08	94.10
Stressed Grass	98.29	98.11	**99.26**	98.00
Synthetic Grass	99.74	**100.00**	**100.00**	99.94
Tree	93.32	96.93	97.18	**97.50**
Soil	95.44	**100.00**	**100.00**	99.44
Water	99.70	**100.00**	**100.00**	99.97
Residential	92.22	94.46	97.39	**98.00**
Commercial	**91.94**	81.98	88.87	86.70
Road	**94.16**	86.14	90.18	90.80
Highway	95.44	97.22	99.07	**99.36**
Railway	94.42	97.57	**99.83**	99.50
Parking Lot 1	**92.95**	88.10	93.21	92.00
Parking Lot 2	**100.00**	98.60	98.37	97.00
Tennis Court	97.67	**100.00**	**100.00**	**100.00**
Running Track	**99.14**	98.71	97.74	97.70
OA(%)	94.66	94.48	96.18	**96.20**
AA(%)	95.34	95.49	**96.81**	96.75
Kappa(%)	94.21	95.71	**96.43**	96.40
Honghu\Methods	MambaHSI	Branch1	Branch2	Ours
Red Roof	**96.47**	96.18	95.11	95.89
Road	95.43	94.90	98.62	**98.91**
Bare Soil	**94.66**	91.91	89.79	91.39
Cotton	96.96	96.96	98.55	**98.73**
Cotton Firewood	96.20	92.67	99.06	**99.14**
Rape	93.90	94.99	94.97	**95.75**
Chinese Cabbage	86.60	88.14	87.72	**95.75**
Pakchoi	**96.08**	92.83	89.96	89.73
Cabbage	94.78	96.46	**98.93**	94.05
Tuber Msatard	90.45	88.15	96.51	**98.45**
Brassica Parachinensis	91.96	87.95	89.19	**96.14**
Brassica Chinensis	87.40	92.37	94.11	**94.44**
Small Brassica Chinensis	89.12	89.47	87.56	**96.88**
Lactuca Sativa	94.01	**94.14**	94.04	90.84
Celtuce	98.84	99.79	**100.00**	94.12
Film Covered Lettuce	96.83	98.85	97.90	**99.35**
Romaine Lettuce	**98.60**	97.61	93.94	98.01
Carrot	96.72	96.66	**97.45**	96.63
White Radish	91.30	90.24	**93.70**	93.50
Garlic Sprout	99.18	**98.96**	97.74	97.71
Broad Bean	99.91	**100.00**	**100.00**	**100.00**
Tree	**99.54**	97.25	96.70	96.58
OA(%)	94.58	94.52	95.43	**96.25**
AA(%)	94.77	94.39	95.07	**96.11**
Kappa(%)	94.00	95.24	94.29	**96.67**
Hanchuan\Methods	MambaHSI	Branch1	Branch2	Ours
Strawberry	89.72	92.76	92.41	**95.05**
Cowpea	81.37	81.79	**89.70**	87.44
Soybean	94.88	93.49	93.37	**97.57**
Sorghum	98.46	**99.53**	99.42	99.10
Water Spinach	99.54	99.57	99.83	**99.91**
Watermelon	**76.99**	57.29	65.08	72.45
Greens	92.61	77.04	87.89	**92.65**
Trees	70.73	50.79	**79.52**	75.23
Grass	**89.01**	80.68	82.94	85.33
Red Roof	94.44	96.55	**97.92**	97.87
Gray Roof	89.56	93.36	**96.31**	92.35
Plastic	88.83	86.01	91.32	**92.72**
Bare Soil	78.69	**84.97**	76.59	80.01
Road	89.68	93.16	93.37	**94.04**
Bright Object	94.42	90.97	96.62	**96.78**
Water	98.28	**99.10**	98.54	98.55
OA(%)	90.21	89.34	92.35	**92.73**
AA(%)	89.20	86.07	90.05	**91.06**
Kappa(%)	89.07	82.67	87.33	**90.44**

modules, obtains excellent performance. The final framework demonstrates superior performance compared with other structures, showcasing the advantages of the proposed LGSFA-Net framework.

b) Ablation on three learnable factors of ESSFF module: To systematically evaluate the impact of the weighting coefficients L_a, L_b, and L_c in the ESSFF module on model performance, we designed a series of ablation experiments. By setting different fixed numerical combinations of L_a, L_b, and L_c in (8), we constructed model variants with varying branch preference weights to investigate the contribution of each branch to the final performance.

The experiment established three distinct weight configuration patterns. The first category is the single-branch dominance mode, which includes retaining only the F_{M} branch ($L_a = 1$, $L_b = 0$, and $L_c = 0$), only the F_{IM} branch ($L_a = 0$, $L_b = 1$, and $L_c = 0$), and only the fusion branch SA(F_{M},F_{IM}) ($L_a = 0$, $L_b = 0$, and $L_c = 1$). The second category is the balanced fusion mode, which involves equally weighting the outputs of the F_{M} and F_{IM} branches ($L_a = 0.5$, $L_b = 0.5$, and $L_c = 0$). The third category is the branch preference mode, specifically including F_{M} branch preference ($L_a = 0.6$, $L_b = 0.2$, and $L_c = 0.2$), F_{IM} branch preference ($L_a = 0.2$, $L_b = 0.6$, and $L_c = 0.2$), and fusion branch SA($F_{\mathrm{M}}, F_{\mathrm{IM}}$) preference ($L_a = 0.2$, $L_b = 0.2$, and $L_c = 0.6$).

As shown in Table V, the experimental results demonstrate that the fusion of F_{M} and F_{IM} branches yields better performance than any single-branch mode. The three-branch fusion strategy further improves stability compared with the average fusion mode. In terms of branch preference, emphasizing either the F_{M} or the fusion branch SA(F_{M}, F_{IM}) leads to superior results over emphasizing the F_{IM} branch alone. Most importantly, the learnable weighting mechanism with adaptive coefficients L_a, L_b, and L_c enables the proposed method to achieve the best performance, demonstrating its clear advantage.

c) Ablation on FCL module: To evaluate the encoder in feature-relation learning, the ablation experiments are done to compare the proposed IMamba with the local encoder and channel encoder. The experimental results are shown in Fig. 10. The experimental results indicate that a model containing only the local encoder performs better compared with a model containing only the channel encoder. When comparing the performance of the model combining both the local encoder and the channel encoder, the joint model demonstrates superior performance. Furthermore, the local encoder has a more significant impact on the performance of the IMamba module, which is particularly evident in the PaviaU dataset.

E. Visualization and Analysis

To further validate the advantages of the proposed model, feature visualization on the four benchmarking datasets is conducted. As shown in Figs. 11–14, we present the ground truth of four benchmark datasets, the feature visualization of the MambaHSI model, the feature visualization of the network model composed of Branch1 or Branch2, and the feature visualization of the LGSFA-Net model proposed in this article. The area in the white boxes is the samples that can perform the advantages of the proposed methods. As shown in

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TABLE VII
COMPARED TRAINING AND TESTING TIME OF THE CONTRASTING METHODS ON PAVIAU DATASET

Metrics	SVM	RF	DMSGer	GiGCN	FullyContNet	CLOLN	Spectralformer	GSC-ViT	MambaHSI	Ourss
$T_{training}$ (s)	0.01	0.29	576.88	556.31	612.15	38.43	94.19	62.21	274.54	497.32
$T_{testing}$ (s)	3.52	1.96	0.09	3.78	0.09	46.46	62.43	49.44	0.03	0.49

TABLE VIII
GFLOPS AND PARAMETERS OF LGSFA-NET ON PAVIAU

Methods	GFlops(G)	Parameters(M)
Only Mamba	7.598	0.164
Mamba+ESSFF	10.43	0.226
Only IMamba	140.043	2.015
IMamba+ESSFF	142.875	2.077
LGSFA-Net	148.964	2.210

Fig. 11–14, the comparison results within the white box in the figure can qualitatively demonstrate the superior performance of the proposed method to some extent. Taking the recognition of trees in the white box in Fig. 14 as an example, when compared with the ground truth, we can observe that the network composed of the IMamba module performs better in identifying the tree category compared with the network composed of the Mamba module. Furthermore, the LGSFA-Net network designed by combining Mamba and IMamba surpasses the prediction results of networks independently formed by the IMamba block and the Mamba block, as well as the recognition performance of networks solely composed of the Mamba module. Compared with the MambaHSI method, the proposed approach demonstrates higher prediction accuracy, enabling better identification of more trees.

To quantitatively analyze the visualization results, a comparative analysis of MambaHSI, Branch1, Branch2, and the proposed LGSFA-Net (Ours) is presented in Table VI. The results demonstrate that LGSFA-Net achieves superior performance in several classes across different datasets. On the PaviaU dataset, LGSFA-Net shows improved accuracy in the categories of Trees, Bare Soil, and Bricks, with increases of 7.13%, 14.45%, and 3.26%, respectively, compared with MambaHSI. For the Houston dataset, LGSFA-Net performs notably well in Tree, Residential, Highway, and Tennis Court. In addition, it exhibits strong classification results in categories such as Road, Cotton, Cotton Firewood, Rape, and Chinese Cabbage on the Honghu dataset. In particular, LGSFA-Net shows significant performance advantages in Chinese Cabbage (7.61% higher), Tuber Mustard (1.94% higher), Brassica Parachinensis (4.18% higher), Small Brassica Chinensis, and Film Covered Lettuce (7.41% higher) compared with MambaHSI, Branch1, and Branch2. On the Hanchuan dataset, LGSFA-Net also delivers considerable accuracy improvements in Strawberry and Soybean, with gains ranging from 2.64% to 5.33% and 2.69% to 4.20%, respectively.

F. Complexity Analysis

Experimental results indicate that the Mamba module itself exhibits relatively low parameter and computational requirements. The introduction of the ESSFF module led to an increase of 0.062M parameters and 2.832 GFLOPs in computational cost compared with models employing only Mamba or IMamba. The final LGSFA-Net architecture contains 2.21M parameters and requires 48.964 GFLOPs of computation. Analysis reveals that the IMamba module constitutes the primary source of both parameters and computational overhead in LGSFA-Net. It integrates convolutional and depthwise separable convolutional operations into the Mamba block, wherein the parameters predominantly arise from the convolutional layers, while the computational cost is largely due to the proposed FCL operations. The involvement of matrix multiplications in these operations significantly contributes to the elevated computational complexity.

As for the model complexity, the training time and testing time are tested for comparison with the main contrasting methods on the PaviaU dataset. As shown in Table VII, it suggests that the proposed LGSFA-Net's training time and testing time are competitive. It behaves more than MambaHSI, but less than the other contrasting methods. According to the results in Tables I–IV, and Table VII, LGSFA-Net performs better performance with little increasing training and testing time, compared with that of MambaHSI.

In addition, as shown in Table VIII, we comprehensively analyze the computational complexity and parameter size across different module configurations, including Mambaonly, IMamba-only, Mamba + ESSFF, IMamba + ESSFF, and the full LGSFA-Net (Mamba + IMamba + ESSFF). The IMamba module introduces significant computational overhead (142.875 GFLOPs and 2.077M parameters) compared with the baseline Mamba implementation, primarily due to its sophisticated feature relationship modeling. However, the ESSFF module demonstrates remarkable efficiency, adding only 10.43 GFLOPs (0.226M parameters) when combined with either module. This computational analysis reveals a clear tradeoff: while the complete LGSFA-Net (IMamba + ESSFF) reaches 142.875 GFLOPs, it delivers a 3.21% OA improvement on PaviaU compared with Mambaonly (148.964 versus 142.875 GFLOPs for comparable accuracy metrics). The 4.2% reduction in computation with superior performance suggests our architecture effectively balances complexity and capability. Further optimization of the IMamba's feature modeling operations could potentially reduce its current 78% contribution to the total computation.

V. CONCLUDING REMARKS

A LGSFA-Net is proposed in this article, which can achieve local and global spectral feature learning by integrating local convolutional inductive biases and the global Mamba block. In LGSFA-Net, a Mamba block with feature interaction learning is introduced, taking into account the correlations between local features, especially focusing on learning the relationships between pixel-level and region-level features in HSIs. It demonstrates high performance in HSI classification tasks, showcasing the potential of the hybrid Mamba framework for

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feature representation. In addition, during the research process, limitations of the framework have been identified, including its inability to directly handle large-scale image interpretation tasks for RS scene classification datasets. Therefore, exploring a hybrid Mamba framework suitable for large-scale RS image interpretation remains one of our future research directions. In addition, the FCL mechanism in IMamba increases GFLOPs, making it an interesting challenge to reduce computational costs and explore lightweight FCL methods.

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